

QTM 220 Final Practice Problems

Problem 1

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ drawn with replacement from a population with mean μ and standard deviation σ .

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n+2} \sum_{i=1}^n Y_i$

Part A

What is $E[\tilde{Y}]$?

$$\begin{aligned} E[\tilde{Y}] &= E\left[\frac{1}{n+2} \sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} E\left[\sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n+2} \sum_{i=1}^n E[Y_i] \end{aligned}$$

by linearity of expectations

$$= \frac{1}{n+2} \sum_{i=1}^n E[Y]$$

by IID

$$= \frac{1}{n+2} \sum_{i=1}^n \mu$$

by definition

$$= \frac{1}{n+2} n\mu$$

$$= \frac{n}{n+2} \mu$$

Part B

How does $E[\tilde{Y}]$ compare to $E[\bar{Y}]$? (where $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$)

similar steps show that $E[\bar{Y}] = \mu$. $\bar{Y}$ is unbiased, $\tilde{Y}$ is biased, but $\lim_{n \rightarrow \infty} \frac{n}{n+2} = 1$ so bias will decrease as n increases.

Part C

Showing your work, calculate $Var[\tilde{Y}]$.

$$Var[\tilde{Y}] = V[\frac{1}{n+2} \sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 V[\sum_{i=1}^n Y_i]$$

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n V[Y]$$

because IID, covariances are 0

$$= (\frac{1}{n+2})^2 \sum_{i=1}^n \sigma^2$$

by definition

$$= (\frac{1}{n+2})^2 n\sigma^2$$

$$= \frac{n}{(n+2)^2} \sigma^2$$

Part D

How does $\text{Var}[\tilde{Y}]$ compare to $\text{Var}[\bar{Y}]$?

Similar steps show that $V[\tilde{Y}] = \frac{\sigma^2}{n}$

$\frac{1}{n} > \frac{n}{(n+2)^2}$ so our $\tilde{Y}$ estimator has a lower variance than the mean. But the estimator is biased, so we probably wouldn't want to use it over the mean.

Problem 2

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. We run this code.

```
mod<- lm(y~x, data=df)
summary(mod)
```

Call:

```
lm(formula = y ~ x, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.2190	-0.8373	-0.0159	0.8350	4.1442

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.58927	0.14861	-3.965	0.000102 ***
x	0.33179	0.04095	8.102	5.48e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.31 on 198 degrees of freedom

Multiple R-squared: 0.249, Adjusted R-squared: 0.2452

F-statistic: 65.64 on 1 and 198 DF, p-value: 5.481e-14

```
B=10000
beta_0<- rep(NA,B)
beta_1<- rep(NA,B)
for( b in 1:B ){
```

```

bootind = sample(1:n,size=n,replace=T)
dfboot = df[bootind,]
mod<- lm(y~x, data=dfboot)
beta_0[b] = mod$coef[1]
beta_1[b] = mod$coef[2]
}

```

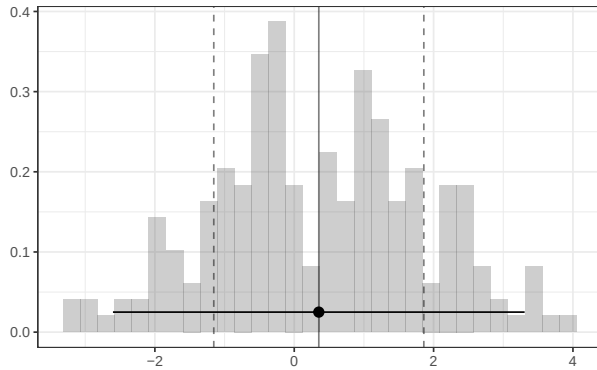


Figure 1: Histogram of the vector Y's elements

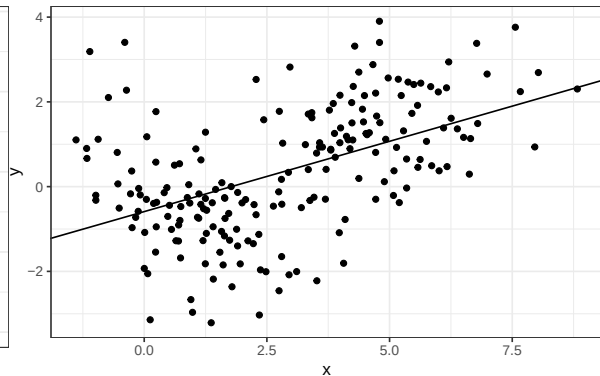


Figure 2: Scatterplot of X and Y

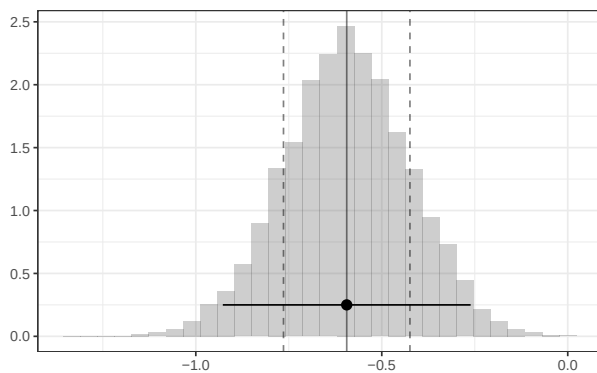


Figure 3: Histogram of the beta_0's elements

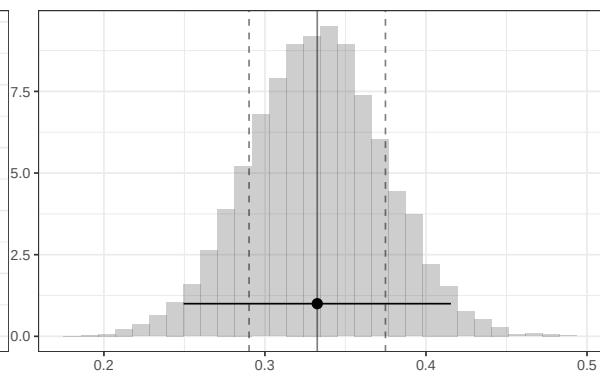


Figure 4: Histogram of the beta_1's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean. It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a ‘b’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation of Y. It’s the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod<- lm(y~x, data=df)
mod$coef[1]
```

The intercept of the simple regression of y on x. It is the intercept of the line on the scatterplot

Part d

What does this code calculate? State it in words and mark it with a ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(beta_1)
```

The bootstrap sampling distribution’s standard deviation for the coefficient on x from the simple regression (slope). Estimated standard error. It’s the distance between the solid vertical and either dashed vertical line on the beta_1 sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a ‘e’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mod$coef[2] + sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X)%*%solve(t(X)%*%X)))[2]
```

```
mod$coef[2] - sqrt(diag(solve(t(X)%*%X)%*%t(X)%*%diag(resid^2)%*%X)%*%solve(t(X)%*%X)))[2]
```

A 95% confidence interval — based on normal approximation — for the least squares slope from the simple regression of y on x for the population from which this sample was drawn. This is indicated by the horizontal segment (with a dot in the middle) on the sampling distribution histogram (`beta_1`).

Part f

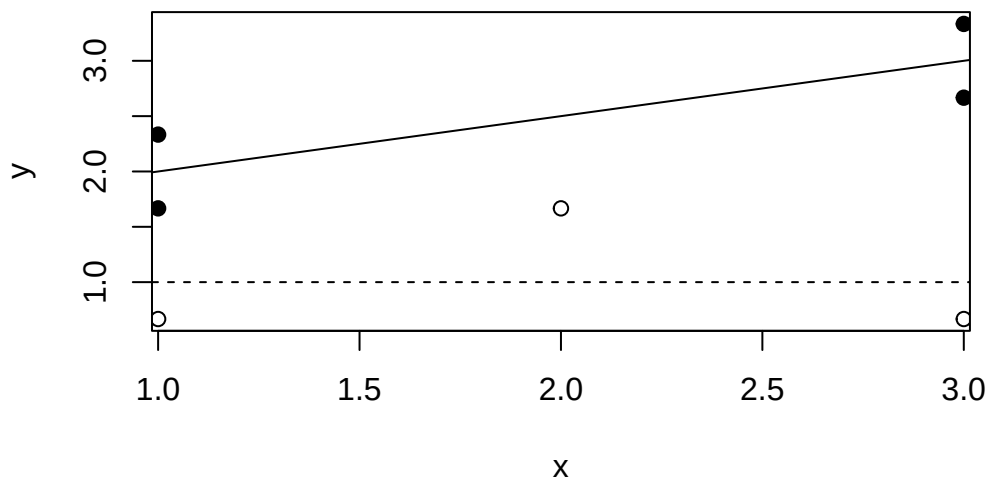
How could you calculate roughly the same thing as in part e using `sd(beta_1)`

We can use the standard deviation of the bootstrap sampling distribution as our estimate of standard error.

```
mod$coef[2] - 1.96 * sd(beta_1)
mod$coef[2] + 1.96 * sd(beta_1)
```

Problem 3

Consider the following data set with treatment d , covariate x , and outcome y , which has been fit with a linear interactive model. Note that the treated units with $d = 1$ are denoted as solid points, while the control units with $d = 0$ are denoted with open circles. The regression line for the treated units is solid, while the regression line for the control units is dashed.



Part A

What is the $\widehat{ATE}$ based on this model and data? Show your calculations.

Part B

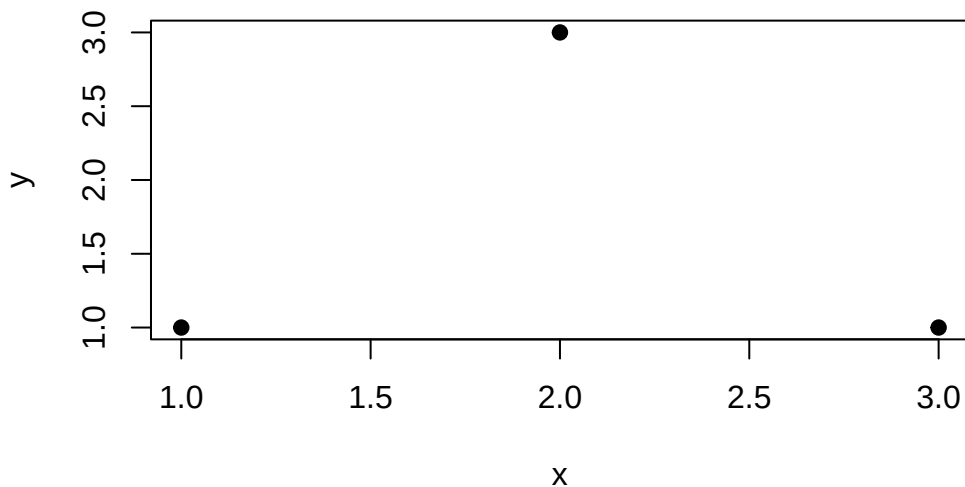
What is the $\widehat{ATT}$ based on this model and data? Show your calculations.

Part B

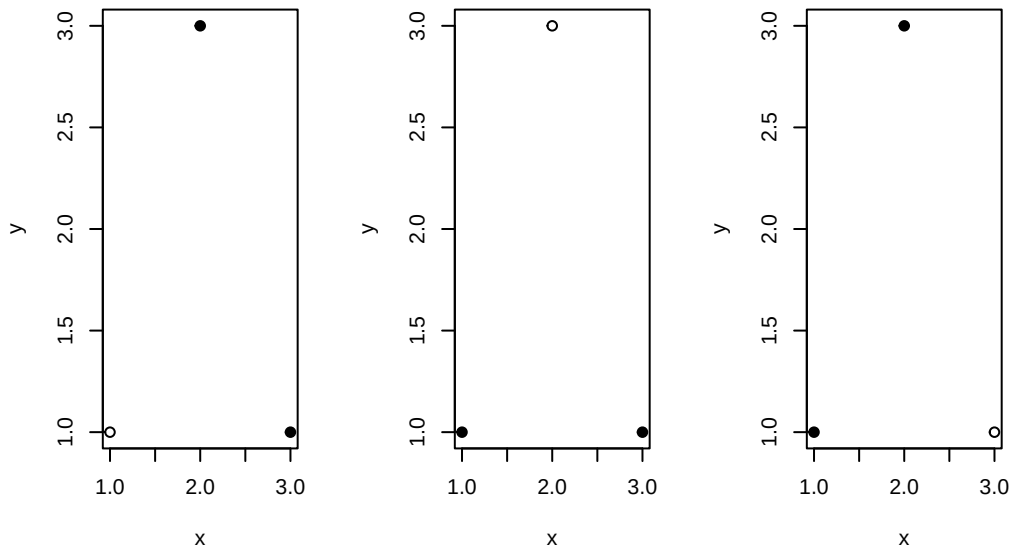
What is the $\widehat{ATC}$ based on this model and data? Show your calculations.

Problem 4

Consider the following data with three observations.



Our goal is to use leave one out cross-validation to choose between a linear regression model and a constant model for this data. Below we have plotted these data three times. In the left panel, the left-most observation has been left out (denoted by an open circle). In the middle panel, the middle observation has been left out. In the right panel the right observation has been left out.



Part A

On the figure above, draw a constant model (i.e., flat line) for each of the three plots based on the non-left-out data.

Part B

On the figure above, draw the residuals from the constant models to the left-out observation for each of the three plots.

Part C

On the figure above, draw a linear regression line (i.e., line that doesn't have to be flat) for each of the three plots based on the non-left-out data.

Part D

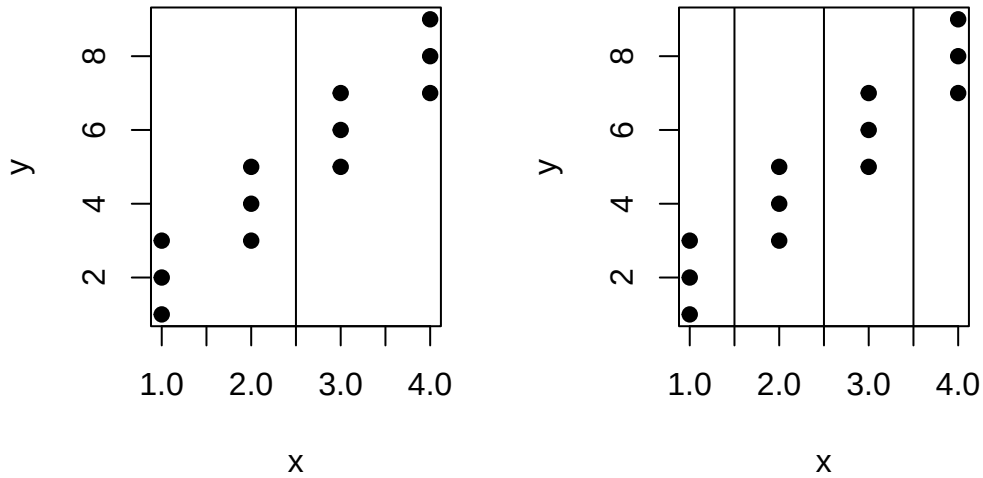
On the figure above, draw the residuals from the linear models to the left-out observation for each of the three plots.

Part E

Based on the residuals you calculated in parts B and D, which model is preferred?

Problem 5

Consider the data below. In the left panel, it has been partitioned into two bins. In the right panel, it has been partitioned into four bins.



Part A

In the left panel, draw piece-wise constant models for the two bins.

Part B

Calculate the estimated standard error for each of the two bins in the left panel. Use the table below to organize your calculations.

Observation i	y_i	bin constant	deviation	deviation ²
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

Part C

In the right panel, draw piece-wise constant models for the four bins.

Part D

Calculate the estimated standard error for each of the four bins in the right panel. Use the table below to organize your calculations.

Observation i	y_i	bin constant	deviation	deviation ²
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

Part E

Comment on the difference(s) between the two bin and four bin models.

Problem 5

Part A

Suppose we want β_1 from $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + e_i$ but we run the regression $lm(y \sim x)$ without z . We have de-meaned our variables x_i and z_i such that $\bar{x} = \sum_{i=1}^n x_i = 0$ and $\bar{z} = \sum_{i=1}^n z_i = 0$. Under what condition(s) will the coefficient on x from $lm(y \sim x)$, which is $\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$, be equal to β_1 ? Derive what your coefficient on x will be in terms of β s and justify your steps.

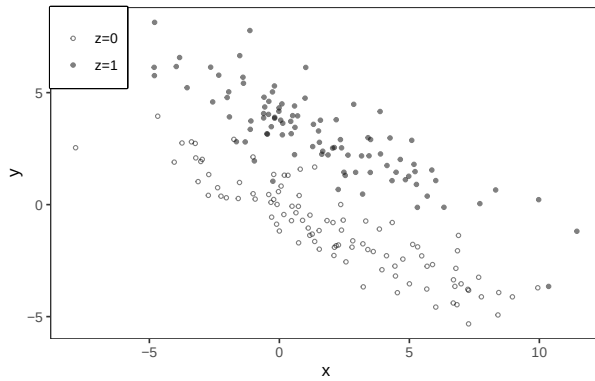


Figure 5: Scatterplot 1

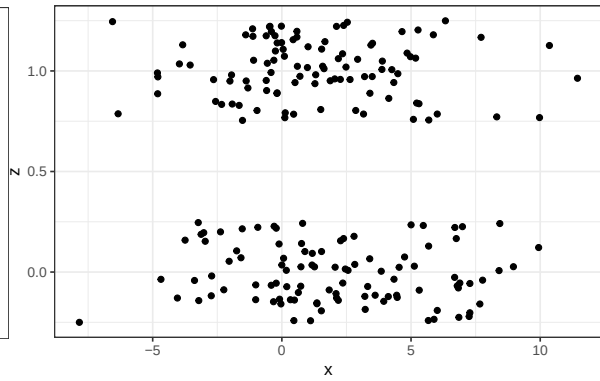


Figure 6: Scatterplot 2 (Jittered)

Part B

We have collected y , x , and a binary variable z . The scatter plots below show our data.

Do you expect the coefficient on x from the simple regression of y on x ($lm(y \sim x)$) to be:

- Positive
- Negative
- Zero

Part C

We then run the regression $lm(y \sim x + z)$ which gives us the following output:

Call:

```
lm(formula = y ~ x + z, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.9710	-0.6505	-0.0814	0.6073	3.3079

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.00805	0.10772	0.075	0.941
x	-0.50992	0.02014	-25.322	<2e-16 ***
z	3.87919	0.14127	27.459	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9935 on 197 degrees of freedom
Multiple R-squared: 0.8877, Adjusted R-squared: 0.8865
F-statistic: 778.4 on 2 and 197 DF, p-value: < 2.2e-16

And we run the regression $lm(z \sim x)$ which gives us the following output:

Call:

```
lm(formula = z ~ x, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.64150	-0.49917	-0.00015	0.48622	0.64447

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.52525	0.03928	13.371	<2e-16 ***
x	-0.01483	0.01008	-1.472	0.143

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4998 on 198 degrees of freedom
Multiple R-squared: 0.01083, Adjusted R-squared: 0.005832
F-statistic: 2.167 on 1 and 198 DF, p-value: 0.1426

How would you use these outputs to calculate the coefficient on x from the simple regression $lm(y \sim x)$? Explain your steps and define any notation you use. Plug in the estimates from your output (you can leave it as an equation).

Part D

How do you expect the standard errors of the simple regression of $y \sim x$ to differ from the standard errors of the multiple regression $y \sim x + z$? Why?